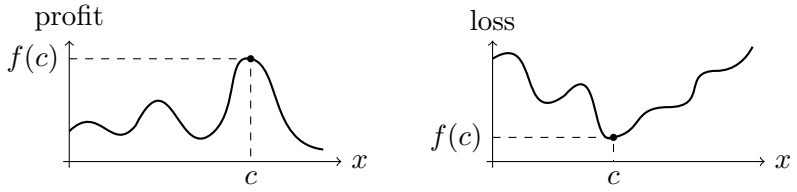
Global and local extrema

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Where does a profit function attain its largest value? Where does a loss function attain its smallest value?



Definition (global extrema). Let D be the domain of f and let $c \in D$. The value $f(c)$ is a

$$\begin{aligned} \textit{global maximum} \quad & \text{if } f(c) \geq f(x) \quad \text{for every } x \in D, \\ \textit{global minimum} \quad & \text{if } f(c) \leq f(x) \quad \text{for every } x \in D. \end{aligned}$$

Either is a *global extreme value*, attained at c.

Theorem (Extreme Value Theorem). If f is continuous on $[a, b]$, then f attains both a global minimum and a global maximum on $[a, b]$.

Note. One can prove this using Bolzano–Weierstrass from analysis, or a result from topology: the continuous image of a compact set is compact. For this class, the conclusion is that continuous functions map closed intervals to closed intervals:

$$f([a, b]) = [m, M],$$

where m and M are the minimum and maximum. The Intermediate Value Theorem supplies every value between them.

Definition (local extrema). The number $f(c)$ is a

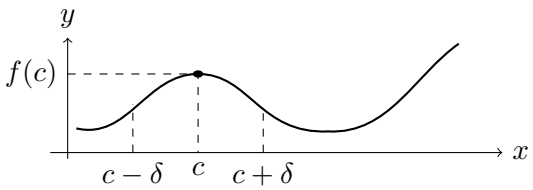
$$\begin{aligned} \textit{local maximum of } f \quad & \text{if } f(c) \geq f(x) \text{ when } x \text{ is near } c, \\ \textit{local minimum of } f \quad & \text{if } f(c) \leq f(x) \text{ when } x \text{ is near } c. \end{aligned}$$

Here x is *near* c if we restrict x to some open interval containing c, considering only points in the domain.

Note. More precisely, a local maximum means that some $\delta > 0$ satisfies

$$f(x) \leq f(c) \qquad \text{for every } x \in D \cap (c - \delta, c + \delta).$$

For a local minimum, reverse the inequality. At an included endpoint, compare only on the side in the domain.



The value $f(c)$ is a local maximum. Farther to the right, f takes larger values.

Finding global extrema

Definition (critical number). A number c in the domain of f is a *critical number* if $f'(c) = 0$ or $f'(c)$ does not exist. The corresponding output $f(c)$ is a *critical value*.

Theorem (Fermat: extrema $\implies$ critical). If f has a local maximum or minimum at c, then c is at the boundary of the domain or is a critical number of f.

Proof. It suffices to consider an interior point c. If $f'(c)$ does not exist, c is critical by definition. Suppose $f'(c)$ exists and $f(c)$ is a local maximum. For all sufficiently small h, we have $f(c+h) \le f(c)$, so the numerator $f(c+h) - f(c)$ is nonpositive.

If $h > 0$, dividing by h preserves the inequality; if $h < 0$, it reverses it. Thus

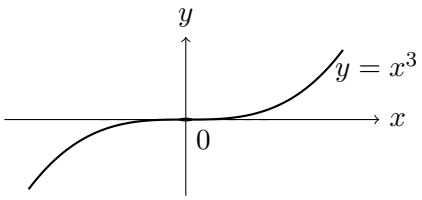
$$\frac{f(c+h) - f(c)}{h} \le 0 \quad (h > 0), \qquad \frac{f(c+h) - f(c)}{h} \ge 0 \quad (h < 0).$$

Since $f'(c)$ exists, the right and left limits both equal $f'(c)$. Taking these limits gives

$$f'(c) \le 0 \qquad \text{and} \qquad f'(c) \ge 0.$$

Hence $f'(c) = 0$. At a local minimum, the inequalities reverse and give the same conclusion. In either case, c is critical. □

Remark (critical $\implies$ extrema is false). For $f(x) = x^3$, we have $f'(0) = 0$, so 0 is critical. But $x^3 < 0$ to the left of 0 and $x^3 > 0$ to the right. Thus $f(0)$ is neither a local maximum nor a local minimum.



Closed-interval method. To find the global extrema of a continuous function f on $[a, b]$:

1. List the endpoints a and b.
2. Find all critical numbers in (a, b).
3. Evaluate f at every listed point. The largest and smallest outputs are the global maximum and minimum.

Example 1. Find the global extrema of $f(x) = x + 2$ on $[-1, 1]$. Since $f'(x) = 1$, there are no interior critical numbers. Compare

$$f(-1) = 1, \qquad f(1) = 3.$$

The global minimum is 1 at $x = -1$ and the global maximum is 3 at $x = 1$.

Example 2. Find the global extrema of $f(x) = x^3(4 - x)$ on $[-1, 4]$. We have

$$f'(x) = 12x^2 - 4x^3 = 4x^2(3 - x).$$

Compare both endpoints and the two interior critical numbers:

$$f(-1) = -5, \qquad f(0) = 0, \qquad f(3) = 27, \qquad f(4) = 0.$$

The global minimum is -5 at $x = -1$ and the global maximum is 27 at $x = 3$.

Examples

Example 3. Find the global extrema of $f(x) = 1 - x^2$ on $[-1, 3]$. Since $f'(x) = -2x$, the only interior critical number is 0. The candidate values are

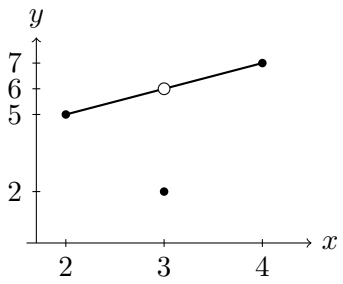
$$f(-1) = 0, \qquad f(0) = 1, \qquad f(3) = -8.$$

The maximum is 1 at $x = 0$ and the minimum is -8 at $x = 3$.

Example 4. Find the global extrema on $[2, 4]$ of

$$f(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & x \neq 3, \\ 2, & x = 3. \end{cases}$$

Away from 3, $f(x) = x + 3$ and $f'(x) = 1$. At 3, f is discontinuous and $f'(3)$ does not exist. The maximum is $f(4) = 7$ and the minimum is $f(3) = 2$.



Example 5. Find the global extrema of $f(x) = \sin x$ on $[-\pi/2, \pi]$. The only interior zero of $f'(x) = \cos x$ is $\pi/2$. Compare

$$f(-\pi/2) = -1, \qquad f(\pi/2) = 1, \qquad f(\pi) = 0.$$

The minimum is -1 at $x = -\pi/2$ and the maximum is 1 at $x = \pi/2$.

Example 6. Find the global extrema of $f(x) = 2 + |x|$ on $[-1, 2]$. The graph has slope -1 to the left of 0 and 1 to the right, so 0 is a critical number. From $f(-1) = 3$, $f(0) = 2$, and $f(2) = 4$, the minimum is 2 at $x = 0$ and the maximum is 4 at $x = 2$.

Example 7. For $f(x) = 2 + (x - 5)^3$ on $[4, 7]$, is 5 a critical number? Is $f(5)$ a local extremum? Since $f'(x) = 3(x - 5)^2$, we have $f'(5) = 0$. Yet $f(x) < 2$ to the left of 5 and $f(x) > 2$ to the right. Thus 5 is a critical number and $f(5)$ is neither a local maximum nor a local minimum.